Rollup Economics: Sequencing, Pricing and Validation

Akaki Mamageishvili

Offchain

October 30, 2026; CBER CtCe

Introduction to Rollups

Ethereum (L1) needs scaling, which (optimistic) rollups (L2) provide

- maintain a separate state from Ethereum
- move execution off the chain

Effectively, a separate chain that Ethereum is aware of

Rollup Mechanics

State update:

- user sends a transaction to a sequencer
- sequencer receives transactions
- sequencer passes transaction to execution stage in some order
- with enough transactions sequencer posts a batch/blob to Ethereum
- these transactions become canonical for updating the rollup state

Sequencer trusted to include transactions

- delayed inbox of transactions

Transaction Ordering

There is value in the ordering:

- blockchains use auctions to get some value back

Ethereum *Proposer Builder Separation (PBS)*:

- first (second) price auction where builders participate

Builders:

- run auctions among searchers
- find optimal knapsack solutions with constraints
- have long term relationships with searchers/wallets/applications

Ordering in Rollups

The sequencer acts as a block builder in L1, with the difference that:

- it is single party and has a fixed ordering rule
- maintains a private mempool
- more convenient for (financial) applications

Most common ordering rule:

- transactions are sorted in decreasing priority fees per gas
- corresponding to priority gas auction (PGA)
- originally a default ordering rule on L1

Currently, priority fees on L1 are usually used for inclusion

FCFS

First-come first-served

- established transaction ordering rule in traditional finance

Few rollups use FCFS:

- Robinhood
- Arbitrum until April, 2025

Arbitrum used TimeBoost between April, 2025 and September, 2026

- speed advantage (bump) for one party

FCFS vs. PGA

FCFS:

- pros: "fair"; fastest execution resulting into best prices;
- cons: latency competition and resources that are lost from protocol; spam;

PGA:

- pros: protocol captures some share of the value; fair in some sense;
- cons: slows down execution; hard to find tradeoff between speed and revenue;

TimeBoost

In between FCFS and auctioning advantage:

- a protocol sells a time advantage in the FCFS policy to one party
- a permissionless, second-price auction with a reserve price
- if no bidder bids more than the reserve price, transaction ordering policy remains pure FCFS
- the time advantage equals 200ms and lasts 1 minute for the winner

The ordering rule is enforced by the sequencer:

- executes regular transactions with a 200ms delay
- transactions coming from the TimeBoost winner are executed on their arrival; effectively, the sequencer merges these two FCFS queues

Just-in-Time vs. Ahead-of-Time

PBS and PGA have one commonality: both are "Just-in-Time" (JIT):

- bidders can bid immediately before the content of the block is determined

Alternative to JIT: "Ahead-of-Time" (AOT) auctions

- a sizable time gap between the auction and the determination of the value

In L1, these auctions have been discussed under the term "slot auctions":

- the builder only bids on the right to propose the next block
- no commitment to any specific block content at the time of bidding
- the builder determines block content only (12 seconds) later

AOT vs. JIT

To compare the relative performance of AOT and JIT auctions:

- assess how well bids match the value that bidders get from the right they acquire

In JIT auctions, the performance is just a question of competition:

- as long as enough bidders compete in the auction the paid bids should be close to the value that the winning bidders get

For AOT auctions, besides competition, there is a problem of prediction

Empirical Analysis

"Just-In-Time Resale in an Ahead-Of-Time Auction for Faster Execution" (with Öz, Schlegel, Taslimi, 2026)



Figure: Paper link

Prediction

Bidders need to predict the future value that they will get:

- if they are able to predict it well (and there is sufficient competition), the auction will yield efficient outcomes
- if they do not, the auction will generally be more inefficient than a JIT auction

Outline of Insights

Compared cumulative fixed time markout of fast lane trades over the TimeBoost interval to bids for the fast lane:

- markout is not fast lane value, but the best measure we can observe

Comparison allows to assess how well bids predict future extracted value

- bids are a noisy predictor of extracted value
- correlation between winning bids and markouts is weak across bidders

Relative order of the most frequent bidder performance remains the same

- across different settings
- together with their absolute profit estimates

Coordination Phase

The auction transformed from competition to coordination:

- one bidder always winning
- the winner resells time advantage to two previous competing bidders
- reselling details not visible on-chain
- activity/profits of searchers remain similar to competitive phase
- revenue of the auction declines drastically

Insights for Ethereum

The likely performance of slot auctions:

- estimating immediate future arbitrage value is hard
- auction revenue and efficiency may suffer from switching from block to slot auctions
- casts some doubts on the idea of slot auctions as a market structure for block building

Pricing

L2s need to price not only their own resources, but also

- pass L1 batch/blob posting fees to its users
- manage a delay of data posting
- avoid spam

The last issue is solved by setting non-trivial lower bound on gas fees

- all major L2s have it
- on Ethereum, it is delegated to block builders

Beyond Congestion

Chain (L1 or L2) pricing so far handles congestion, through EIP 1559:

- increase price/gas fees when total usage is above target
- decrease gas fees when total usage is below target

The mechanism is:

- agnostic to revenue
- neutral to usage patterns as it tracks real resource usage

It does not have to only capture congestion:

- for revenue maximization, demand curve estimates are needed

Elasticity

"Price Elasticity of Gas Demand on L1 and L2: Evidence from Ethereum and Arbitrum" (with Anchuri, 2026)

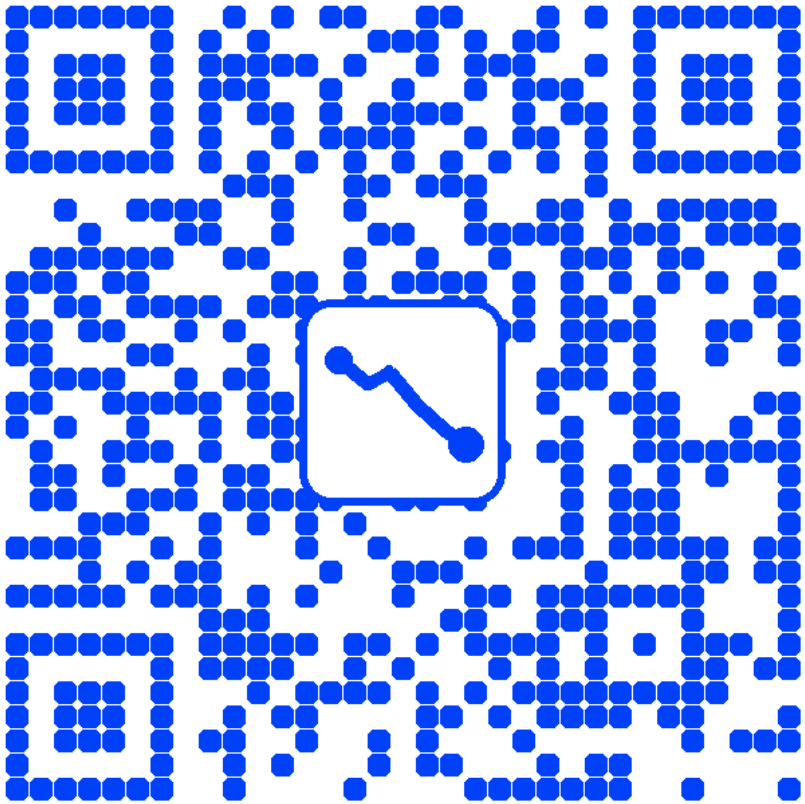


Figure: Paper link

Data

Ethereum's single gas and Arbitrum's multi-resource consumption:

- computation
- historyGrowth
- storageAccessRead
- storageAccessWrite
- storageGrowth
- l2Calldata
- refund

Insights

Difficulty to infer causality:

- gas fees rise because of rising demand by EIP-1559

Results:

- clustering users based on their gas/resource consumption patterns
- both chains are mostly inelastic
- Arbitrum is 6x more elastic than Ethereum

Some resources on Arbitrum are more elastic than others:

- can be explained by probabilistic backruns

Validation

(Properly designed) rollups keep security by:

- posting the transaction relevant call-(blob-)data to Ethereum
- some rollup specific mechanism for fraud proofs
- one of the validators posts a current state of a rollup to base layer
- anyone can challenge
- correct claim wins
- wrong asserter gets punished

Verifier's dilemma

No pure equilibrium with right claim and checking, if the latter is costly:

- validator posts right claim
- verifier does not check since checking is not free
- validator does not need to post the right claim

Incentivize validators to *actually* check results

- only rational players and economic incentives

Validation and Attention

"One Node, Two Roles: Simultaneous Contests for Validation and Attention in Rollups" (with Anchuri, Berger, and Campanelli, 2026)



Figure: Paper link

Modeling

Framework:

- dividing service providers into validators and attention players
- entry cost to pay attention

Each validation or attention service incurs costs

- can not be reduced by delegating

The players decide:

- which service to provide
- how many Sybils to launch for each service

Modeled and analyzed through simultaneous Tullock contests

Insights

With non-decreasing total rewards:

- sufficient condition on costs to have no-Sybil equilibrium

With arbitrary (decreasing) total rewards:

- unconditional on entry costs or costs of running service(s) to have no-Sybil equilibrium

Related to discussion on issuance curve in PoS protocols

Thank you!

Questions?